

- 256.** Clearly, 2525 is not divisible by any of the numbers 7, 11 and 13.

The smallest 3-digit prime number is 101.

$$\begin{array}{r} 25 \\ 101 \overline{)2525} \\ \underline{202} \\ 505 \\ \underline{505} \\ 0 \end{array} \quad \begin{array}{r} 32 \\ 101 \overline{)3232} \\ \underline{303} \\ 202 \\ \underline{202} \\ 0 \end{array}$$

Hence (d) is true.

- 257.** Numbers like 2525, 3636 etc. are divisible by 101.
Numbers like 256256, 678678 etc. are divisible by 1001.
Numbers like 32163216, 43754375 etc. are divisible by 10001 and so on.
- 258.** 10^n has $(n + 1)$ digits. Then, 9 will appear n times in $(10^n - 1)$.
So, sum of digits in $(10^n - 1) = 9n$.
 $\therefore 9n = 4707 \Rightarrow n = \frac{4707}{9} = 523$.
- 259.** We know that $(x^n - a^n)$ is always divisible by $(x - a)$ for all values of n .
- 260.** The product of 8 consecutive numbers is divisible by each one of $8!, 7!, 6!, 5!, 4!, 3!$ and $2!$
- 261.** We know that $(x^n - a^n)$ is divisible by $(x - a)$, when n is even and $(x^n - a^n)$ is divisible by $(x + a)$, when n is even.
 $\therefore (10^n - 1)$ is divisible by $(10 - 1) = 9$, when n is even.
And, $(10^n - 1)$ is divisible by $(10 + 1) = 11$, when n is even.
 $\therefore$ Statements 2 and 4 are correct.
- 262.** Putting $n = 1$, we get $(3^4n - 4^3n) = (3^4 - 4^3) = (81 - 64) = 17$, which is divisible by 17.
- 263.** Let the odd natural number be $(2n + 1)$.
 $n = 1$ gives $(2n + 1)^2 = (2 + 1)^2 = 9$.
This when divided by 8 gives 1 as remainder.
 $n = 2$ gives $(2n + 1)^2 = 25$.
This when divided by 8 gives 1 as remainder and so on.
- 264.** Required number $= (2^5 - 2) = (32 - 2) = 30$.
- 265.** Two consecutive even integers are $2n$ and $(2n + 2)$.
 $\therefore (2n + 2)^2 - (2n)^2 = (4n^2 + 4 + 8n - 4n^2) = 4 + 8n = 4(1 + 2n)$, which is always divisible by 4.
- 266.** Two consecutive odd integers are $(2m + 1)$ and $(2m + 3)$.
 $\therefore (2m + 3)^2 - (2m + 1)^2 = (2m + 3 + 2m + 1)(2m + 3 - 2m - 1) = (4m + 4) \times 2 = 8(m + 1)$, which is always divisible by 8.
- 267.** The smallest 4-digit number is 1000.
This when divided by 7 leaves 6 as remainder.
 $\therefore$ 1001 is the smallest 4-digit number exactly divisible by 7.
- 268.** On dividing 1056 by 23, we get 21 as remainder.
 $\therefore$ Required number to be added $= (23 - 21) = 2$.

$$\begin{array}{r} 23 \overline{)1056} (45 \\ \underline{92} \\ 136 \\ \underline{115} \\ 21 \end{array}$$

- 269.** On dividing 8567 by 4, the remainder is 3.
To make it divisible by 4, we must add 1 to it.
- 270.** The least 6-digit number $= 100000$.
Required number $= 100000 + (349 - 186) = 100000 + 163 = 100163$.

$$\begin{array}{r} 349 \overline{)100000} (286 \\ \underline{698} \\ 3020 \\ \underline{2792} \\ 2280 \\ \underline{2094} \\ 186 \end{array}$$

- 271.** The greatest 5-digit number $= 99999$.
On dividing 99999 by 279, we get 117 as remainder.
 $\therefore$ Required number $= (99999 - 117) = 99882$.

$$\begin{array}{r} 279 \overline{)99999} (358 \\ \underline{837} \\ 1629 \\ \underline{1395} \\ 2349 \\ \underline{2232} \\ 117 \end{array}$$

- 272.** The greatest 6-digit number is 999999.
Required number to be added $= (327 - 33) = 294$.

$$\begin{array}{r} 327 \overline{)999999} (3058 \\ \underline{981} \\ 1899 \\ \underline{1635} \\ 2649 \\ \underline{2616} \\ 33 \end{array}$$

- 273.** On dividing 5000 by 73, we get
Required number $= 5000 + (73 - 36) = 5037$.

$$\begin{array}{r} 73 \overline{)5000} (68 \\ \underline{438} \\ 620 \\ \underline{584} \\ 36 \end{array}$$

- 274.** On dividing 58701 by 567, we get 300 as remainder.
 $\therefore$ Required number $= 58701 + (567 - 300) = 58701 + 267 = 58968$.

$$\begin{array}{r} 567 \overline{)58701} (103 \\ \underline{567} \\ 2001 \\ \underline{1701} \\ 300 \end{array}$$

- 275.** On dividing 8112 by 99, we get 93 as remainder.
So, the required number to be subtracted is 93.

$$\begin{array}{r} 99 \overline{)8112} (81 \\ \underline{792} \\ 192 \\ \underline{99} \\ 93 \end{array}$$

- 276.** On dividing 803642 by 11, we get 4 as remainder.
Required number to be added $= (11 - 4) = 7$.
- 277.** On dividing 1111 by 99, the quotient is 11 and the remainder is 22.
Hence, the required number is 11.

$$\begin{array}{r} 99 \overline{)1111} (11 \\ \underline{99} \\ 121 \\ \underline{99} \\ 22 \end{array}$$

- 278.** $66 = 11 \times 6$
In order to get a number divisible by 18, the above product must be multiplied by 3.
Hence 66 must be multiplied by 3.
- 279.** The smallest 6-digit number is 100000. On dividing 100000 by 111, we get 100 as remainder.
So, the number to be added $= (111 - 100) = 11$.
Hence, the required number $= 100011$.

$$\begin{array}{r} 111 \overline{)100000} (900 \\ \underline{999} \\ 100 \end{array}$$